7.3 Proving That a Quadrilateral Is a Parallelogram

Name _____ Period ____ Date _____

Geometry 7.3 (30-minute plan) · Student sheet

I can…

- name the theorem that proves a quadrilateral is a parallelogram.
- find values of variables that make a quadrilateral a parallelogram.
- use slope, distance, or midpoints to show a quadrilateral in the coordinate plane is a parallelogram.

Vocabulary · fill in the missing word

Converse — the statement made by switching the _____ and the conclusion

Parallelogram Opposite Sides Converse — both pairs of opposite sides _____ → parallelogram

Parallelogram Opposite Angles Converse — both pairs of opposite angles _____ → parallelogram

Opposite Sides Parallel and Congruent Theorem — one _____ of opposite sides both congruent and parallel → parallelogram

Parallelogram Diagonals Converse — diagonals _____ each other → parallelogram

Part 1 · Notes and examples (fill in each note, then work the example)

Definition of a parallelogram: A quadrilateral is a parallelogram if both pairs of opposite sides are _____ (definition).

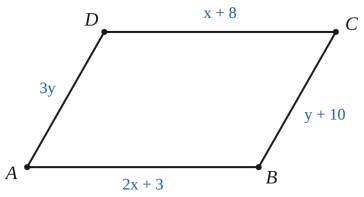
Parallelogram Opposite Sides Converse: A quadrilateral is a parallelogram if both pairs of opposite sides are _____.

Parallelogram Opposite Angles Converse: A quadrilateral is a parallelogram if both pairs of opposite angles are _____.

Example 1

For what values of x and y is ABCD a parallelogram?

AB = 2x + 3, DC = x + 8, AD = 3y, BC = y + 10



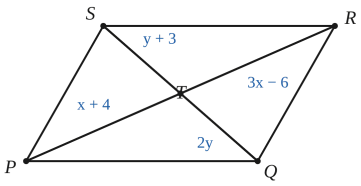
Opposite Sides Parallel and Congruent Theorem: A quadrilateral is a parallelogram if one pair of opposite sides is _____ and _____.

Parallelogram Diagonals Converse: A quadrilateral is a parallelogram if the diagonals _____ each other.

Coordinates: use _____ (parallel), _____ (congruent), or _____ of the diagonals (bisect).

Example 2

The diagonals of quadrilateral PQRS meet at T. PT = x + 4, TR = 3x − 6, QT = 2y, and TS = y + 3. For what values of x and y is PQRS a parallelogram?



Part 2 · Partner practice

Partner practice

In quadrilateral ABCD, m∠A = (4x − 10)° and m∠C = (2x + 30)°. For what value of x could ABCD be a parallelogram (with m∠B = m∠D)?